Contour Generator Inequalities

Verification, Sharp Refinements, and Value for 3D Reconstruction

Independent mathematical assessment and consolidated results

6 November 2026

Bottom line. The proposed anisotropy correction is a valid, strictly sharper[1] bound on the *full spatial rim average*. It is *not* currently a demonstrated improvement over established 3D reconstruction methods, since practical images provide visible projected silhouettes rather than complete 3D rims, and the correction uses curvatures of the reconstructed shape. For image-accessible *convex-hull* silhouettes, Topping's 2026 mean-width theorem supplies a sharper, already-established inequality. The strongest immediate use is as a *geometric consistency certificate*, not as a stand-alone reconstruction algorithm.

Proven main refinement:

$$\boxed{\mathcal{L}_{\mathrm{rim}}^2 \le \frac{A}{2}(\mathcal{W} - \pi\chi) - \frac{A}{64}\mathcal{D}, \qquad \mathcal{D} = \int_\Sigma \frac{(\kappa_1^2 - \kappa_2^2)^2}{\kappa_1^2 + \kappa_2^2}\, dA.} \tag{1}$$

The quotient in $\mathcal{D}$ is taken to be zero when both curvatures vanish. No bibliographic originality is claimed for this refinement; it is an independently checked mathematical deduction.

1 Setting, observables, and why the distinction matters

Let $\Sigma \subset \mathbb{R}^3$ be a nonempty smooth compact embedded orientable surface without boundary, unless otherwise specified. It may be nonconvex or disconnected. Set

$$H = \tfrac{1}{2}(\kappa_1 + \kappa_2), \quad K = \kappa_1\kappa_2, \quad A = \int_\Sigma dA, \quad \mathcal{W} = \int_\Sigma H^2\, dA, \quad \chi = \chi(\Sigma), \quad q = \kappa_1^2 + \kappa_2^2.$$

For generic $u \in \mathbb{S}^2$, the spatial rim is $\Gamma_u = \{p \in \Sigma : \langle N(p), u\rangle = 0\}$, and

$$\mathcal{L}_{\mathrm{rim}} = \frac{1}{4\pi}\int_{\mathbb{S}^2} \mathcal{H}^1(\Gamma_u)\, d\omega(u).$$

Define Π_u to be orthogonal projection onto $u^\perp$. The *projected rim length with multiplicity* averages the integral of $|\Pi_u T|$ over the tangent T to each rim branch, and is denoted $\mathcal{L}_{\mathrm{proj}}$. The average length of the actual *visible* projected silhouette is $\mathcal{L}_{\mathrm{vis}}$, counting only image-set contours actually visible. The mean *convex-hull projection perimeter* is

$$\mathcal{L}_{\mathrm{hull}} = \frac{1}{4\pi}\int_{\mathbb{S}^2} \mathrm{Per}(\Pi_u \operatorname{conv}\Sigma)\, d\omega(u).$$

They are different quantities. In the usual orthographic opaque-surface model, $\mathcal{L}_{\mathrm{vis}} \le \mathcal{L}_{\mathrm{proj}} \le \mathcal{L}_{\mathrm{rim}}$, but neither $\mathcal{L}_{\mathrm{hull}} = \mathcal{L}_{\mathrm{rim}}$ nor $\mathcal{L}_{\mathrm{hull}} = \mathcal{L}_{\mathrm{vis}}$ holds in general. The convex hull of an observed silhouette can be used to obtain $\mathcal{L}_{\mathrm{hull}}$, assuming calibrated orthographic views and complete object masks. Perspective data require additional modelling.

[1] Strict whenever the anisotropy functional defined below is positive; not strict on every surface.

2 Exact spatial identity and original sharp bound

Theorem 2.1 (Incidence-coarea identity)**.** *With $B = -dN$ and $C_p = \{u \in \mathbb{S}^2 : u \perp N(p)\}$, define the parametrized curvature-ellipse perimeter*

$$P_B(p) = \int_{C_p} |B_p u| \, d\mathcal{H}^1(u) = \int_0^{2\pi} \sqrt{\kappa_1^2 \cos^2\theta + \kappa_2^2 \sin^2\theta} \, d\theta.$$

Then $\mathcal{L}_{\text{rim}} = \dfrac{1}{4\pi} \displaystyle\int_\Sigma P_B \, dA$. Degenerate ellipses are counted by their parametrized limiting perimeter.

Proof. Consider $F(p,u) = \langle N(p), u\rangle$ on $\Sigma \times \mathbb{S}^2$, and $M = F^{-1}(0)$. At each $(p,u) \in M$, $N(p) \in T_u\mathbb{S}^2$ and $d_u F[N(p)] = 1$, so M is a smooth compact three-manifold. Sard's theorem applied to the projection $\pi_u : M \to \mathbb{S}^2$ yields smooth one-dimensional rim fibers for almost every u. On M, $\nabla_p F = -B_p u$, $\nabla_u F = N$. Writing $a = |B_p u|$, the two-dimensional projection Jacobians are $J_2\pi_p = (1+a^2)^{-1/2}$ and $J_2\pi_u = a(1+a^2)^{-1/2}$. Coarea first over π_u and then over π_p gives

$$\int_{\mathbb{S}^2} \mathcal{H}^1(\Gamma_u) \, d\omega(u) = \int_M J_2\pi_u \, d\mathcal{H}^3 = \int_\Sigma \int_{C_p} |B_p u| \, d\mathcal{H}^1(u) \, dA.$$

There is no division by the potentially vanishing quantity a. Divide by 4π. □

Lemma 2.2 (Ellipse comparison)**.** *For any two real principal curvatures,*

$$4\sqrt{q} \le P_B \le \pi\sqrt{2q}. \tag{2}$$

The upper equality holds precisely when $|\kappa_1| = |\kappa_2|$; the lower equality holds when one principal curvature is zero (or both vanish).

Proof. For the upper estimate apply Jensen to the concave square root of $\kappa_1^2\cos^2\theta + \kappa_2^2\sin^2\theta$. For the lower bound divide by $\sqrt{q}$, let $t = \kappa_1^2/q \in [0,1]$, and observe that $\int_0^{2\pi} \sqrt{t\cos^2\theta + (1-t)\sin^2\theta}\, d\theta$ is concave in t, hence at least the common endpoint value 4. The equality claims follow from strict concavity away from degeneracies. □

Theorem 2.3 (Original sharp Willmore-type bound)**.**

$$\boxed{\mathcal{L}_{\text{rim}}^2 \le \frac{A}{2}(\mathcal{W} - \pi\chi).} \tag{3}$$

The constant $1/2$ is sharp. For connected surfaces, equality holds exactly for round spheres; in general it holds precisely for finite disjoint unions of equal-radius round spheres.

Proof. Theorems above, Cauchy–Schwarz, and Gauss–Bonnet give

$$\mathcal{L}_{\text{rim}}^2 \le \frac{A}{8}\int_\Sigma q \, dA = \frac{A}{8}\left(4\mathcal{W} - 2\int_\Sigma K \, dA\right) = \frac{A}{2}(\mathcal{W} - \pi\chi).$$

A sphere of radius r has $\mathcal{L}_{\text{rim}} = 2\pi r$, $A = 4\pi r^2$, $\mathcal{W} = 4\pi$, $\chi = 2$. Conversely, equality forces $|\kappa_1| = |\kappa_2|$ pointwise and $|B|$ constant globally. Nonzero opposite principal curvatures on any component would give a compact minimal surface in $\mathbb{R}^3$, which is impossible by harmonicity of the coordinate functions. Therefore all components are totally umbilic round spheres, with common radius from the global constant $|B|$. □

Corollary 2.4 (Total curvature comparison)**.**

$$\boxed{\frac{1}{\pi}\int_\Sigma |B|\,dA \le \mathcal{L}_{\mathrm{rim}} \le \frac{1}{2\sqrt{2}}\int_\Sigma |B|\,dA, \qquad |B| = \sqrt{q}.} \tag{4}$$

Both constants are optimal for the pointwise ellipse estimates; the lower equality is not asserted for a closed surface.

3 Proved anisotropy refinement: all genera

Lemma 3.1 (Second-order ellipse deficit)**.** *For $q > 0$, let $t = (\kappa_1^2 - \kappa_2^2)/q \in [-1, 1]$. Then*

$$\boxed{P_B^2 \le 2\pi^2\left(q - \frac{(\kappa_1^2 - \kappa_2^2)^2}{8q}\right).} \tag{5}$$

The same holds by continuity when $q = 0$.

Proof. For uniformly distributed $\theta \in [0, 2\pi]$, write $Z = \cos 2\theta$. Then Z and $-Z$ have the same law, $\mathbb{E}Z^2 = 1/2$, and

$$P_B = 2\pi\sqrt{q/2}\,\mathbb{E}\sqrt{1 + tZ}.$$

For $|y| \le 1$,

$$\left(\frac{\sqrt{1+y} + \sqrt{1-y}}{2}\right)^2 = \frac{1 + \sqrt{1-y^2}}{2} \le 1 - \frac{y^2}{4}.$$

By symmetry and Jensen, $\mathbb{E}\sqrt{1+tZ} \le \mathbb{E}\sqrt{1 - t^2Z^2/4} \le \sqrt{1 - t^2/8}$. Squaring proves the assertion. □

Theorem 3.2 (Strengthened Willmore-type inequality)**.** *Set $\mathcal{D} = \int_\Sigma (\kappa_1^2 - \kappa_2^2)^2/q\,dA$, zero at $q = 0$. Then*

$$\boxed{\mathcal{L}_{\mathrm{rim}}^2 \le \frac{A}{2}(\mathcal{W} - \pi\chi) - \frac{A}{64}\mathcal{D}.} \tag{6}$$

The inequality holds in every genus, for nonconvex and disconnected surfaces, and is an equality on round spheres (and equal-radius unions of spheres). The correction is nonnegative and positive unless $|\kappa_1| = |\kappa_2|$ almost everywhere.

Proof. By Theorem 2.1, Cauchy–Schwarz, and Lemma 3.1,

$$\mathcal{L}_{\mathrm{rim}}^2 \le \frac{A}{16\pi^2}\int_\Sigma P_B^2\,dA \le \frac{A}{8}\int_\Sigma q\,dA - \frac{A}{64}\mathcal{D}.$$

The identity $\int_\Sigma q\,dA = 4(\mathcal{W} - \pi\chi)$ completes the proof. □

Remark 3.3 (Earlier preliminary coefficient and its limitation). An earlier candidate $31/2048$ in place of $1/64$ is valid but weaker, since $31/2048 < 1/64$. The improved coefficient comes from the exact symmetry argument above. $\mathcal{D}$ measures anisotropy of *curvature magnitudes*; it *does not* measure total non-umbilicity: $\kappa_1 = -\kappa_2 \neq 0$ gives $\mathcal{D}$'s integrand zero although the point is not umbilic. This matters for reconstruction interpretations.

Corollary 3.4 (No explicit Euler term)**.** *Let $B^\circ = B - HI$. Then*

$$\boxed{\mathcal{L}_{\mathrm{rim}}^2 \le \frac{A}{4}\mathcal{W} + \frac{A}{8}\int_\Sigma |B^\circ|^2\,dA - \frac{A}{64}\mathcal{D}.} \tag{7}$$

This is algebraically equivalent to (6), since $\int |B^\circ|^2 = 2\mathcal{W} - 4\pi\chi$; it does not remove topology in substance, only encodes it geometrically.

4 Projected contours and visible silhouettes

Theorem 4.1 (Projected rim formula, counted with multiplicity)**.** *The exact averaged orthographic projected rim length is*

$$\boxed{\mathcal{L}_{\text{proj}} = \frac{1}{4\pi}\int_{\Sigma}\int_{0}^{2\pi}\left|\kappa_1\cos^2\theta + \kappa_2\sin^2\theta\right|\,d\theta\,dA.} \tag{8}$$

Consequently

$$\frac{1}{2}\int_{\Sigma}|H|\,dA \le \mathcal{L}_{\text{proj}} \le \mathcal{L}_{\text{rim}} \le \sqrt{\frac{A}{2}(\mathcal{W}-\pi\chi) - \frac{A}{64}\mathcal{D}}. \tag{9}$$

If $\kappa_1\kappa_2 \ge 0$ everywhere (in particular on convex surfaces), $\mathcal{L}_{\text{proj}} = \frac{1}{2}\int|H|\,dA$. For ordinary visible silhouettes, $\mathcal{L}_{\text{vis}} \le \mathcal{L}_{\text{proj}}$.

Proof. At a regular rim point, the unit rim tangent is $T = JB_pu/|B_pu|$, with J rotation by 90° in the tangent plane. Therefore $|\Pi_u T| = |\langle B_pu, u\rangle|/|B_pu|$. Weighting the incidence coarea identity by $|\Pi_u T|$ gives $\int_\Sigma\int_{C_p}|\langle B_pu,u\rangle|\,d\mathcal{H}^1(u)\,dA$, which is (8). The first lower bound follows by the triangle inequality, the middle from $|\langle Bu,u\rangle| \le |Bu|$; visible curves are a subset of all projected branches, with repetitions omitted. For same-sign principal curvatures, the absolute value in (8) can be removed pointwise. □

Visibility flag. On nonconvex surfaces, $\mathcal{L}_{\text{proj}}$ includes occluded branches and overlap multiplicity; its exact formula is *not* an exact formula for the observed visible silhouette. A method using only 2D photographs cannot directly evaluate $\mathcal{L}_{\text{rim}}$, $\mathcal{D}$, or $\mathcal{W}$ without estimating 3D geometry.

5 The false proposed improvement and the correct convex bound

Proposition 5.1 (Counterexample to the JPG claim for spatial rims)**.** *The proposed universal inequality $\mathcal{L}^2_{\text{rim}} \le A\mathcal{W}/4$ is false, even among smooth closed convex genus-zero surfaces.*

Proof. First, for the ring torus $R = 2, r = 1$,

$$A = 8\pi^2, \qquad \mathcal{W} = 4\pi^2/\sqrt{3}, \qquad \kappa_1 = 1, \quad \kappa_2 = \cos t/(2+\cos t).$$

Because $P_B \ge \int_0^{2\pi}|\cos\theta|\,d\theta = 4$, we have $\mathcal{L}_{\text{rim}} \ge A/\pi = 8\pi$, while $\sqrt{A\mathcal{W}/4} = \sqrt{8\pi^4/\sqrt{3}} < 8\pi$. Hence the inequality fails exactly, without numerical approximation. Second, take increasingly long cylinders of radius r, length h, smoothly capped by convex ends of bounded size. As $h/r \to \infty$, the cylindrical part dominates: $\mathcal{L}_{\text{rim}} \sim 2h$, $A \sim 2\pi rh$, $\mathcal{W} \sim \pi h/(2r)$, and hence $\mathcal{L}^2_{\text{rim}}/(A\mathcal{W}) \to 4/\pi^2 > 1/4$. These are smooth genus-zero convex counterexamples. □

Theorem 5.2 (Optimal convex spatial-rim bound)**.** *For every smooth closed convex surface,*

$$\boxed{\mathcal{L}^2_{\text{rim}} \le \frac{4}{\pi^2}A\mathcal{W}.} \tag{10}$$

The constant $4/\pi^2$ is optimal in the convex class, approached by elongated smoothly capped cylinders.

Proof. When κ_1, κ_2 have the same sign, $P_B \le \int_0^{2\pi} (|\kappa_1 \cos\theta| + |\kappa_2 \sin\theta|)\, d\theta = 4(|\kappa_1| + |\kappa_2|) = 8|H|$. Then $\mathcal{L}_{\mathrm{rim}} \le (2/\pi) \int |H|\, dA \le (2/\pi)\sqrt{A\mathcal{W}}$. The cylinder family in the preceding proof achieves the constant asymptotically. □

Thus convex surfaces enjoy the combined sharper estimate

$$\boxed{\mathcal{L}_{\mathrm{rim}}^2 \le \min\left\{ \tfrac{A}{2}(\mathcal{W} - 2\pi) - \tfrac{A}{64}\mathcal{D},\ \tfrac{4}{\pi^2} A\mathcal{W} \right\}.}$$

Neither branch dominates on every convex surface: the first is exact on the sphere, while the second is asymptotically exact on long rounded cylinders.

6 Convex-hull projected perimeter: a different, stronger established theorem

For connected smooth closed embedded surfaces, the 2026 theorem of Topping [9] states

$$\int_\Sigma (H_{\mathrm{out}})_+\, dA \ge \frac{1}{2} \int_{\mathbb{S}^2} w_\Sigma(e)\, d\omega(e), \quad w_\Sigma(e) = \max_{p \in \Sigma} \langle p, e \rangle - \min_{p \in \Sigma} \langle p, e \rangle. \tag{11}$$

Here H_{out} is the signed curvature in Toppingś outward-normal convention, positive on a round sphere: if our shape operator is $B = -dN$ with N outward, then $H_{\mathrm{out}} = -H$. Thus $|H_{\mathrm{out}}| = |H|$ and the Willmore energy is unchanged. The theorem is a nontrivial external input; we do not reprove it here. Cauchy's planar perimeter formula, averaged over view directions, yields

$$\mathcal{L}_{\mathrm{hull}} = \frac{1}{4} \int_{\mathbb{S}^2} w_\Sigma(e)\, d\omega(e).$$

Therefore $2\mathcal{L}_{\mathrm{hull}} \le \int (H_{\mathrm{out}})_+ \le \int |H| \le \sqrt{A\mathcal{W}}$, and

$$\boxed{\mathcal{L}_{\mathrm{hull}}^2 \le \frac{A\mathcal{W}}{4}.} \tag{12}$$

The sphere attains equality. Miura's November 2026 preprint [10] extends sharp mean-width statements to certain immersed surfaces; the connected embedded formulation above is independently verified directly against Topping's Theorem 1.4. For convex Σ, the equality $\mathcal{L}_{\mathrm{hull}} = \mathcal{L}_{\mathrm{proj}} = \frac{1}{2} \int H\, dA$ is the classical Minkowski mean-curvature formula. *None of this proves* $\mathcal{L}_{\mathrm{rim}}^2 \le A\mathcal{W}/4$. Convex-hull perimeter and spatial rim length cannot be interchanged.

7 Numerical tests and genus questions

The curvature-ellipse formula was evaluated by high-resolution deterministic quadrature for revolution tori. The columns labelled bounds are *square roots* of the right-hand side, in the same length units as $\mathcal{L}_{\mathrm{rim}}$.

Surface (R, r)	$\mathcal{L}_{\mathrm{rim}}$	original (3)	refined (6)	exact P_B-CS bound
Ring torus $(2, 1)$	28.233659	29.997111	28.804162	28.443808
Ring torus $(\sqrt{2}, 1)$	21.609934	23.474008	22.660458	22.469854
Ring torus $(3, 0.5)$	38.462964	42.169127	39.599351	38.466860

The "exact P_B-CS bound" is $\sqrt{A\int P_B^2\,dA/(16\pi^2)}$, itself a tighter geometric estimate than the anisotropy simplification, but it requires the entire curvature-ellipse field. These numerical checks are independent of, and weaker than, the analytical proofs.

For connected embedded genus-g surfaces consider the *proposed optimization problem*

$$c(g) = \sup_{\text{genus}(\Sigma)=g} \frac{4\mathcal{L}_{\text{rim}}^2}{A\mathcal{W}}. \tag{13}$$

The established examples and original bound yield

$$\frac{16}{\pi^2} \le c(0) \le 2, \qquad 1.7717624\ldots \le c(1) \le 2.$$

The genus-zero lower bound comes from elongated rounded cylinders; genus-one from the $(2,1)$ ring torus. These are *bounds, not evaluations* of $c(0)$ and $c(1)$. A proof of optimal $c(g)$, its extremizers, or a nontrivial stability estimate would be a substantially more ambitious mathematical result. We do not assert that this optimization problem is novel or unstudied without exhaustive bibliographic verification.

8 Comparison with standard Willmore theory and 3D reconstruction

8.1 Which statements compete, and which do not?

Result / method	Input	Reconstruction interpretation
Classical $\mathcal{W} \ge 4\pi$ [6]	Willmore energy	Universal bending-energy baseline; equality on a round sphere.
Marques–Neves torus [7]	Genus one; $\mathcal{W} \ge 2\pi^2$	Strong topology-sensitive result, not an image-fitting algorithm.
Discrete Willmore flow [8]	Mesh curvature and gradients	Directly supports smoothing, restoration, and hole filling.
Differentiable reconstruction [12, 13]	Views, images, appearance, geometry	Directly optimizes geometry against observations.
Original rim, (3)	Full 3D rim; A, W, χ	Theoretical consistency check; rim is latent in opaque photographs.
Refined rim, (6)	Adds $\mathcal{D}$	Tighter certificate, but needs reconstructed principal curvatures.
Topping hull, (12)	Projected hulls; A, W	Closely linked to image convex-hull perimeter; sharp and already known.

For ordinary multi-view silhouettes, reconstruction is nonunique: hidden concavities cannot in general be recovered from visual-hull methods, even with many views [3, 4]. Modern multi-view surface methods combine photometric/differentiable losses and representation-specific constraints [12, 13, 11]. There is no established reconstruction benchmark in the reviewed literature showing that enforcing (6) improves geometric accuracy, Chamfer distance, silhouette IoU, inference speed, or robustness. Such claims require experiments.

8.2 How to turn the inequality into a testable reconstruction constraint

For a fully sampled directional average $s = \mathcal{L}_{\mathrm{vis}}$, the chain $s \le \mathcal{L}_{\mathrm{rim}}$ implies the *necessary* condition

$$\boxed{\mathcal{W} \ge \pi\chi + \frac{2s^2}{A} + \frac{\mathcal{D}}{32}.} \tag{14}$$

This can flag inconsistent candidate values, and can inspire a soft loss based on a positive-part constraint residual. But it is *not* a sufficient condition for a correct reconstruction and is usually looser than using the full 3D rim. For mean convex-hull perimeters measured directly from silhouettes, Topping's established (12) instead gives

$$\boxed{\mathcal{W} \ge \frac{4\mathcal{L}_{\mathrm{hull}}^2}{A}} \quad \text{(connected embedded surfaces).} \tag{15}$$

The different measurements prevent an unconditional numerical ranking of (14) and (15). A finite set of camera viewpoints does not automatically equal the uniform spherical average; any stochastic approximation needs angular-sampling error analysis. In addition, both constraints require an area estimate, and (14) further needs a topology choice and reliable curvature estimation.

8.3 Evidence-based assessment

- **Mathematical correctness: high.** The incidence-coarea identity, original sharp theorem, anisotropy correction, projected formula, and convex special case have analytic proofs above.
- **Theoretical interest: moderate.** The correction improves the ellipse-perimeter estimate. Publication priority and substantial conceptual novelty remain unestablished.
- **Direct 3D reconstruction value: currently low/unproven.** The full rim is latent, the new deficit requires curvature data, and practical methods already use local curvature/surface regularization and image-fit constraints.
- **Potential utility: real but conditional.** Candidate-shape validation, noise-aware regularization experiments, and calibrated silhouettes for convex or nearly convex objects are plausible settings.
- **Best research direction: empirical and geometric.** Compare a baseline multi-view reconstruction with and without a well-defined contour-consistency penalty; include convex vs nonconvex families, hidden cavities, view subsampling, curvature noise, and topology mistakes. Separately investigate genus-specific optimal $c(g)$ and stability.

Novelty and priority flag. The curvature-ellipse identity and Willmore-type inequality were supplied in the two-page note attributed to *İmamverdi Bədəlov*. The six-page mathematical audit reconstructs proofs and adds projected/equality results. The anisotropy refinement and computational comparisons in this report are additional independently developed deductions. Mathematical correctness does *not* establish that any claim is bibliographically new, and attribution/publication rights must be resolved by human authors. Topping's 2026 mean-width theorem and classical Minkowski identities must be attributed to their authors.

9 Full checklist of established results, rejected claims, and open items

Status	Claim	Basis
Proved	Exact coarea/curvature-ellipse identity; Sard regularity	Theorem 2.1
Proved	$4\sqrt{q} \le P_B \le \pi\sqrt{2q}$	(2)
Proved	Original sharp rim bound; sphere and equal-sphere equality cases	Theorem 2.3
Proved	Two-sided $\int \lvert B\rvert$ comparison	(4)
Proved	Improved $-A\mathcal{D}/64$ correction	Theorem 3.2
Proved	Prior coefficient $31/2048$ is weaker	Coefficients
Equivalent	Trace-free expression with no explicit χ	(7)
Proved	Projected formula; visible $\le$ projected $\le$ spatial	(8)–(9)
Proved	Convex projected formula and optimal spatial $4AW/\pi^2$ bound	Theorem 5.2
Disproved	Universal rim $\mathcal{L}^2_{\mathrm{rim}} \le AW/4$, even smooth convex genus zero	Prop. 5.1
Known result	Hull $\mathcal{L}^2_{\mathrm{hull}} \le AW/4$, from mean-width theorem	Topping 2026
Numerically checked	Three ring tori: $(2,1), (\sqrt{2},1), (3,0.5)$	Section 7
Bounds only	$16/\pi^2 \le c(0) \le 2$; $1.7717624 \le c(1) \le 2$	(13)
Open here	Sharp $c(g)$, stability, genus-only improvement	Research question
Unproven	Better 3D reconstruction using the new constraint	Benchmark needed
Unverified	Originality and scholarly priority	Literature review

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